Hierarchical Concepts in Transformer MLPs: Causal Relevance and Structure

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Abstract

Understanding how large language models (LLMs) represent information is a central challenge in mechanistic interpretability. A key principle in this endeavor is causal relevance, as features that influence model outputs in predictable ways provide stronger evidence of functional importance than those that are merely correlational. Recent work (Shafran et al., 2025) has shown that applying Semi-Nonnegative Matrix Factorization (SNMF) to MLP activations in transformers yields interpretable, hierarchical features, where higher-level concepts emerge from compositions of lower-level sub-concepts.

While these findings offer promising avenues, the causal relevance of hierarchical concepts and their structural organization across layers remain largely unexplored. Prior studies provided illustrative examples but did not rigorously evaluate whether these features causally affect model outputs.

In this paper, we investigate hierarchical features discovered via SNMF along two axes: (i) their causal steerability across hierarchy levels as a proxy for causal relevance, and (ii) the structure of the resulting hierarchies across layers. Through systematic activation steering experiments, we find that features across hierarchy levels in early to mid layers, exhibit a decline in steering performance as the depth increases. Structural analysis further reveals layer-specific organizations, with some layers exhibiting deep, multi-level hierarchies and others showing shallower compositions. These results provide empirical grounding for the causal importance of hierarchical features found in MLP activations, and insights into compositional representations in LLM MLP activations.

1 Introduction

Analyzing how neural networks represent human-interpretable information , remains a key challenge in interpretability research. In Large Language Models (LLMs) such as transformers, internal activations often encode highly distributed representations, making it difficult to isolate meaningful, monosemantic, and functionally relevant features (Gong et al.; Shu et al., 2025). A key goal is to discover representations that are both *interpretable*, corresponding to human-understandable concepts, and *causally relevant*, actively influencing model predictions (Geiger et al., 2025).

A common approach is to use *unsupervised decomposition methods*, which factorize activation patterns into into structured latent features without relying on labeled data (O'Neill et al., 2024; Cunningham et al., 2023; Raghu et al., 2017). Among these methods, *semi-nonnegative matrix factorization* (SNMF) (Lee and Seung, 1999) has proven effective. SNMF decomposes a data matrix $\mathbf{A}$ into a feature matrix $\mathbf{Z}$ and a non-negative coefficient matrix $\mathbf{Y}$:

$$\mathbf{A} \approx \mathbf{ZY}, \quad \mathbf{Z} \in \mathbb{R}^{d\times k}, \quad \mathbf{Y} \in \mathbb{R}^{k\times n}_{\geq 0},$$

where the non-negativity constraint on $\mathbf{Y}$ encourages additive, parts-based representations that often align with human-understandable concepts (Lee and Seung, 1999; Ding et al., 2008). SNMF has been successfully applied across domains, including image analysis (Guo, 2021) and text embeddings (Melas-Kyriazi et al., 2022).

Recent work has applied SNMF to MLP activations in LLMs, producing interpretable features. We refer to these directly extracted features as *level-0 (base) concepts*, with the entries in $\mathbf{Y}$ indicating how strongly each concept is expressed across tokens (Shafran et al., 2025). While (Bricken et al., 2023) identify a *feature-splitting* phenomenon, SNMF instead describe *feature merging*, where these base concepts are recursively composed into higher-level hierarchical features. This recursive decomposition of $\mathbf{Z}$ yields increasingly abstract representations, forming a hierarchy of

features. However, while Shafran et al. confirmed the causal relevance of level-0 features and demonstrated neuron amplification on hierarchical features in a single illustrative example, the causal influence of hierarchical features has not been rigorously evaluated

In this paper, we address this gap by evaluating hierarchical features along two axes: (i) their causal steerability across hierarchy levels, and (ii) the structural organization of hierarchies across layers. Our results show that hierarchical features, exhibit declining steering performance as the depth increases, and that layer-specific differences in hierarchy depth and compositionality reveal structured organization within MLP activations.

We provide the code used in this work, which is publicly available at `https://github.com/ItamarBerger/interp_project_snmf`. The repository is a fork of the codebase from previous work (Shafran et al., 2025) and builds upon it while extending the original implementation.

2 Hierarchical Structure in SNMF-Decomposed Activations

An intriguing observation from SNMF-based analyses is the emergence of *hierarchical structure* in the learned features (Shafran et al., 2025). By recursively applying SNMF to the feature matrix $\mathbf{Z}$-treating its k features as a new activation matrix, higher-level concepts can be expressed as compositions of lower-level sub-concepts. Formally, for an MLP activation matrix $\mathbf{A} \in \mathbb{R}^{d_a \times n}$, this recursive decomposition can be written as:

$$\mathbf{A} \approx Z_L \cdot Y_L \cdot Y_{L-1} \cdots Y_0,$$

where

$$Z_i \coloneqq Z_L \cdot \prod_{j=L}^{i+1} Y_j \in \mathbb{R}^{d_a \times k_i}$$

$$\prod_{j=L}^{i+1} Y_j \coloneqq Y_L Y_{L-1} \cdots Y_{i+1} \in \mathbb{R}_{\geq 0}^{k_i \times n}.$$

In this hierarchy, each feature matrix Z_i represents k_i features at level i, which are compositionally constructed from the lower-level features, which are the columns of Z_{i-1}. For example, fine-grained concepts like "Saturday" and "Sunday" can merge into a broader feature representing "Weekend." Another finding is that semantically related features share core neurons, while certain neurons are exclusive to specific sub-concepts. Prior work (Shafran et al., 2025) demonstrated that such hierarchies naturally emerge in MLP layers, revealing deep compositional structure within activations that reflects meaningful semantic organization.

Despite these findings, critical questions remain. First, the causal relevance of hierarchical features has not been systematically examined. Do hierarchical concepts beyond level 0 exert causal influence comparable to the flat (level-0) features, and does the effectiveness of steering interventions vary with hierarchy depth? Second, what structural properties do these hierarchies exhibit across different layers? Characterizing these structures is essential for understanding how compositional representations are organized.

In this work, we study hierarchical features discovered via SNMF decomposition of MLP activations along two axes: (i) their steering performance across hierarchy levels, and (ii) the structure of the resulting hierarchies across layers of the model.

Our findings suggest that causal relevance, as indicated by steering effectiveness, declines as the depth increases. Further analysis reveals that the hierarchical structure varies across layers: some layers display deep, multi-level hierarchies, while others exhibit shallower compositions.

3 Related Work

3.1 Sparse Autoencoders (SAEs)

A central goal of mechanistic interpretability is to decompose the internal representations of Large Language Models (LLMs) into intelligible components. The current dominant approach for unsupervised feature discovery relies on dictionary learning using Sparse Autoencoders (SAEs) (Bricken et al., 2023; Cunningham et al., 2023). These models are typically trained on residual stream activations to learn directions that correspond to interpretable features (He et al., 2024).

While SAEs have been successful in isolating monosemantic features, they face significant limitations. Primarily, they lack intrinsic interpretability, as their learned features are not explicitly constrained to the model's computational mechanisms (Shafran et al., 2025). Furthermore, recent evaluations suggest that SAE features often struggle in causal tasks, failing to reliably steer model outputs when intervened upon compared to supervised baselines (Huang et al., 2024; Wu et al., 2025).

3.2 Semi-Nonnegative Matrix Factorization (SNMF)

To address the limitations of SAEs, Shafran et al. (Shafran et al., 2025) proposed an alternative approach: decomposing the activations of Multi-Layer Perceptron (MLP) layers directly using Semi-Nonnegative Matrix Factorization (SNMF). Unlike SAEs, which learn features from scratch, SNMF factorizes the activation matrix $A \in \mathbb{R}^{d_a \times n}$ into a sparse feature matrix Z and a non-negative coefficient matrix Y:

$$A \approx ZY \tag{1}$$

This decomposition ensures that the learned features are sparse linear combinations of co-activated neurons, making them "parts-based" and directly grounded in the model's weights (Lee and Seung, 1999; Shafran et al., 2025). Empirical results have demonstrated that SNMF features not only align with human-interpretable concepts but also outperform SAEs in causal steering benchmarks, suggesting they capture functionally relevant units of computation (Shafran et al., 2025).

3.3 Hierarchical Concepts and Steering

Recent theoretical work has begun to formalize how Large Language Models represent complex relationships geometrically. Park et al. (Park et al., 2025) demonstrated that categorical and hierarchical concepts are not merely isolated directions but are often represented as simplices or orthogonal subspaces, where "child" concepts reside within the geometric boundaries of their "parent" concepts.

Building on this theoretical grounding, the original SNMF analysis empirically observed the emergence of such hierarchical structures within MLP activations. By applying SNMF recursively, Shafran et al. showed that high-level concepts (e.g., "Weekdays") are indeed composed of shared "core" neurons (the parent subspace) and distinct "exclusive" neurons that represent sub-concepts (e.g., specific days) (Shafran et al., 2025).

However, prior work has focused primarily on the existence of these hierarchies and the steerability of base-level (Level-0) features. While Park et al. established the geometry and Shafran et al. provided illustrative examples of neuron compositionality, the causal relevance of higher-level hierarchical features remains largely unexplored. Specifically, it has not been rigorously evaluated whether the composite nodes in these concept trees act as effective causal mediators that can steer model behavior comparable to atomic features.

3.4 Our Approach

In this work, we extend the SNMF framework to systematically evaluate the causal validity of hierarchical representations. We investigate hierarchical features along two axes: (i) their causal steerability across different levels of the hierarchy, and (ii) the structural organization of these hierarchies across different model layers. By conducting steering experiments on both base and composite features, we aim to bridge the gap between static interpretability (finding hierarchies) and causal interpretability (verifying their functional role).

4 Experiment

We begin by evaluating the steering of various models using feature vectors extracted from distinct hierarchy levels, adhering to the methodology established by (Shafran et al., 2025). Specifically, we prompt the model with "`<BOS> I think that`" while applying feature steering during inference, and perform grid-search over seven target-values of KL-divergence between a standard forward pass to an intervened forward pass, in order to determine the steering strength (α). Also alternating between negative and positive features to properly attribute features that suppress a concept rather than promote it.

For the SNMF decomposition, we utilize the general concept dataset introduced by (Shafran et al., 2025), which is available in their public repository[1]. Following (Shafran et al., 2025) and building upon prior work (Paulo et al., 2025; Gur-Arieh et al., 2025; Wu et al., 2025), we characterize the extracted features based on their activating inputs (primarily for early-to-mid layers). To reconstruct the concept lineages at each hierarchy level, we identify the activating tokens by computing the product of the coefficient matrices $Y_L \cdots Y_1 Y_0$. This mapping to the input tokens is derived from the hierarchical decomposition objective, which minimizes:

$$\mathcal{L} = \frac{1}{2} \|A - Z_L Y_L \cdots Y_1 Y_0\|_F^2 \tag{2}$$

To generate input-based descriptions for the features using their top activating input tokens, similarly to (Shafran et al., 2025) we use an LLM to generate descriptions based on the top activating tokens. We used Gemini 2.5 Flash (Comanici

[1] https://github.com/ordavid-s/snmf-mlp-decomposition

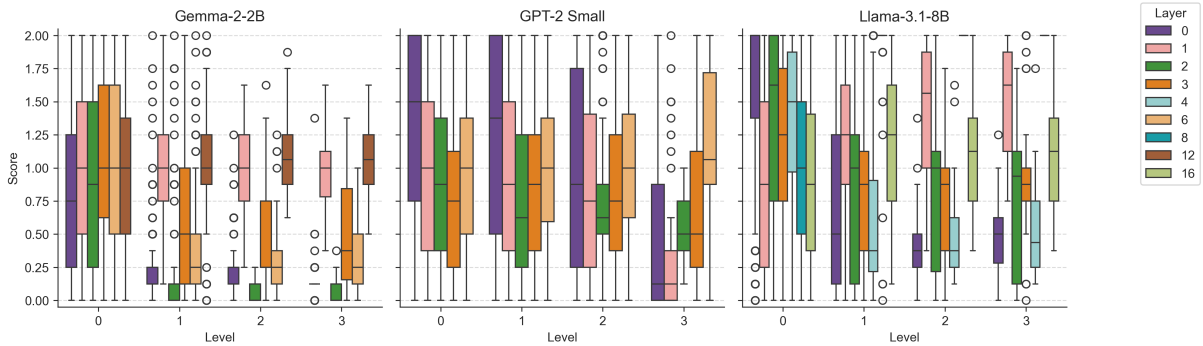

Figure 1: Concept score distribution across SNMF hierarchy levels for Gemma-2-2B (left), GPT-2 Small (center), and Llama-3.1-8B (right) using $k = 400, 200, 100, 50$ for hierarchy levels $0, 1, 2, 3$ respectively.

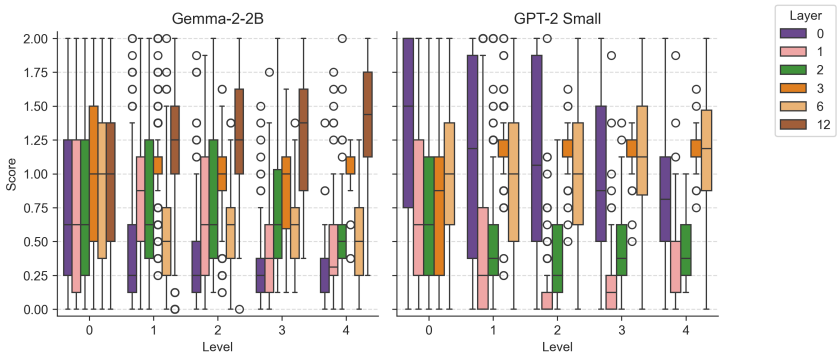

Figure 2: Concept score distribution across SNMF hierarchy levels for Gemma-2-2B (left) and GPT-2 Small (right) using $k = 800, 400, 200, 100, 50$ for hierarchy levels $0, 1, 2, 3, 4$ respectively.

et al., 2025) for generating the input descriptions. The prompt is given in Appendix B. Consistent with (Shafran et al., 2025), we also tried employing output-centric descriptions using vocabulary projections for higher model layers, as suggested by (Gur-Arieh et al., 2025). However, the results we got were really bad. Since hierarchical tree construction relies on input tokens and not on vocabulary projections, we chose to focus our analysis on the early-mid layers of each model. For GPT-2 Small we analyzed the results on layers $0, 1, 2, 3, 6$. For Gemma-2-2B we used layers $0, 1, 2, 3, 6, 12$. For Llama-3.1-8B we used layers $0, 1, 2, 3, 4, 8, 16$.

The evaluation results is done by using Gemini 2.0 Flash as a judge by generating two types of scores: concept score, and fluency score, on each of the steered sentences. For these we used the same prompts as in (Shafran et al., 2025). The "final" score for each steering sentence is the harmonic mean between the fluency score and the concept score.

We conduct steering experiments on multiple layers of Gemma-2-2B (Team et al., 2024), Llama-3.1-8B (Grattafiori et al., 2024), and GPT-2 Small (Radford et al., 2019) using hierarchical ranks of $k = 400, 200, 100, 50$, where $k = 50$ corresponds to the highest (deepest) hierarchical level, and $k = 400$ to the first. Expanding upon (Shafran et al., 2025), we extend the evaluation for Gemma-2-2B and GPT-2 Small to include rank configurations of $k = 800, 400, 200, 100, 50$, a setting not previously tested. This extension aims to assess the impact of higher granularity on steering scores and the diversity of concepts unveiled by the SNMF decomposition.

4.1 Steering Results

The concept scores for the steering experiments are presented in Figure 1 and Figure 2. Final scores are also provided in Figure 3 and Figure 4. We observe that, with minor exceptions, the initial layers (approximately the first 15-25%, varying by model) exhibit a decline in concept scores as the hierarchy depth increases. We hypothesize, though do not formally prove, that this degradation in the early layers stems from their representations lacking sufficient contextualization.

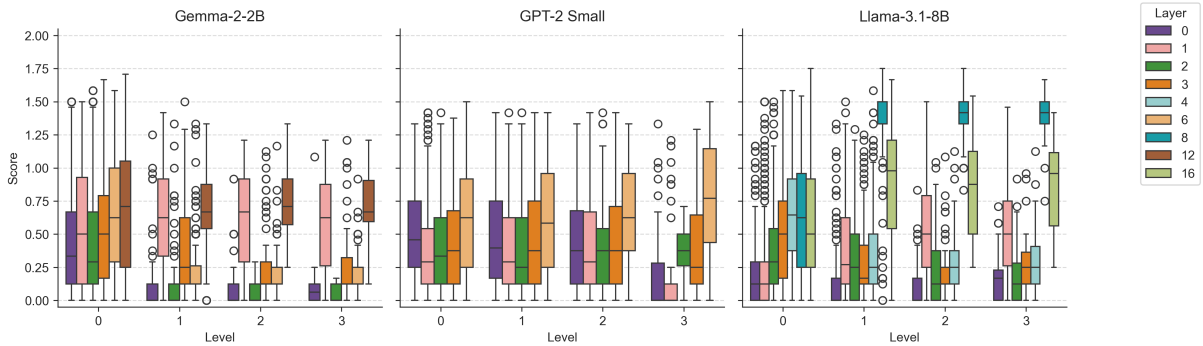

Figure 3: Final score distribution across SNMF hierarchy levels for Gemma-2-2B (left), GPT-2 Small (center), and Llama-3.1-8B (right) using $k = 400, 200, 100, 50$ for hierarchy levels $0, 1, 2, 3$ respectively.

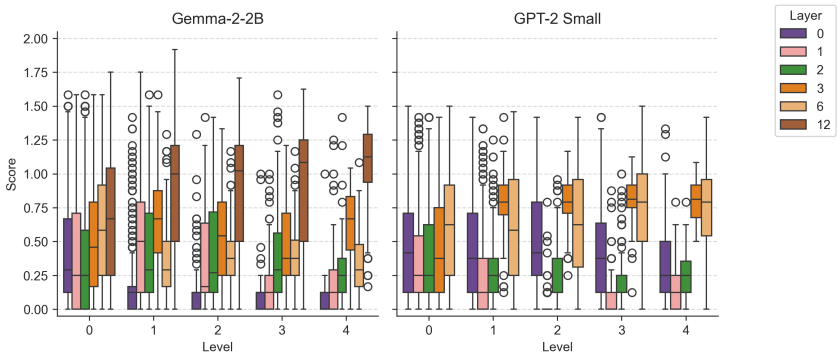

Figure 4: Final score distribution across SNMF hierarchy levels for Gemma-2-2B (left) and GPT-2 Small (right) using $k = 800, 400, 200, 100, 50$ for hierarchy levels $0, 1, 2, 3, 4$ respectively.

Consequently, the SNMF algorithm may encounter optimization challenges in identifying a fitting joint representation across these layers.

Notable exceptions to these trends are observed even within the initial layers. Specifically, in the 4-level decomposition ($k = 400, 200, 100, 50$), layer 1 of Llama-3.1-8B demonstrates the opposite trajectory, where concept scores scale positively with the hierarchy depth. Layer 1 of Gemma-2-2B exhibits consistently elevated scores across all hierarchy levels. A similar anomaly is present in layer 3 of Gemma-2-2B within the 5-level decomposition.

To investigate these deviations, we performed a qualitative analysis of the LLM-generated input descriptions. We found that in the aforementioned cases, the identified features corresponded to broad functional or morphosyntactic categories, such as "Punctuations and Prepositions," "Verbs," or "Actions.".

We categorize these instances as noise; given that functional elements like verbs and punctuation are ubiquitous in model outputs regardless of the target concept, an LLM-based evaluator is likely to assign them high concept scores erroneously. This interpretation is further supported by the final aggregate scores (Figure 3 and Figure 4), which remain significantly lower due to poor fluency metrics. This divergence confirms that while the "concept" (e.g., punctuation) is present, the resulting generations lack the coherence expected from successful steering.

We identified additional anomalies in the steering results, as, for example, within layer 8 of Llama-3.1-8B (Figure 1), where every concept across hierarchy levels 1, 2, and 3 received the maximum concept score of 2. These observations prompted a broader investigation into the relationship between score distributions and the qualitative content of the generated descriptions, independent of the layer of origin.

Our analysis revealed the following trends:

1. For most models and configurations (except for GPT-2 Small with 4-level decomposition): As hierarchy depth increases, feature descriptions tend to converge toward a limited set of recurring patterns, resulting in redundant

conceptual labels across the deeper levels.

2. Higher scores in deeper hierarchy levels correlate with broad morphosyntactic or functional categories (e.g., punctuation or basic parts of speech), as previously noted.

3. An inspection of steering trajectories within individual concept trees across various layers shows that concept scores typically decline as the hierarchy deepens, provided the root concept does not fall into the aforementioned functional categories. This trend persists even in the model's intermediate layers.

A full analysis for these findings can be found in Appendix C.

5 Hierarchical Structure Analysis With Concept Trees

Having introduced hierarchical SNMF structures, we now systematically analyze their organization across layers. One of the findings of (Shafran et al., 2025) suggests that applying recursive decomposition to MLP activations reveals hierarchical structures named *concept trees*. For example, they demonstrated this structure on the week dataset, where it is interpreted as evidence of feature merging.

To construct the trees, we follow their approach, and begin with the highest-level concepts as roots. For each root, we examine concepts from the immediately lower level and connect a lower-level concept as a child if its activation with respect to the parent exceeds the threshold. This procedure is applied recursively across all hierarchy levels, producing directed trees that reflect the compositional organization of MLP features from high-level composites down to base (level-0) concepts.

While recent work (Shafran et al., 2025) primarily uses this procedure to generate illustrative example hierarchies, we extend this approach by performing a systematic analysis of the resulting concept trees. Following the same recursive SNMF-based construction, we connect features across hierarchy levels according to their activation strengths and prune weak connections using a minimal activation threshold (see Section 5.1).

This yields directed trees intended to reflect the hierarchical organization of MLP features across levels.

We apply this tree construction procedure to both GPT-2-small and Gemma-2-2B for different layers, and analyze the structural properties of the resulting hierarchies, such as depth and branching, across layers and hierarchy levels.

5.1 Results

We report the results of the analysis process described above.

The concept trees reveal that hierarchical structure is layer-dependent. Figure 5 shows the average tree depth across layers for both GPT-2-small and Gemma-2-2B. As demonstrated, while some layers consistently produce deeper hierarchies, others remain relatively shallow. Notably, there is a clear trend of increasingly deeper structure as we approach last layers in GPT-2, which may align with previous findings showcasing that more complex features reside in late layers. Additionally, early to middle layers exhibit similar depth patterns across both models, suggesting that hierarchical organization might be a consistent property rather than model-specific noise.

In addition to depth, the breadth of concepts (i.e. number of child sub-concepts) also varies across layers, as illustrated in Figure 6. Some layers show concepts with many children, forming wide, more compositional structures, while others have fewer children, indicating simpler, less compositional representations. These trends reinforce that compositionality is not uniform across the network and is specific to particular layers.

Importantly, while absolute tree depth and branching values are influenced by the pruning threshold used during tree construction, the relative layerwise trends remain robust. In our experiments, GPT-2-small and Gemma-2-2B were analyzed with model-specific thresholds to account for differences in activation scale and sparsity. This ensures that comparisons focus on trends within each model rather than direct cross-model quantitative comparisons. Robustness checks across a range of thresholds (0.1–0.9) confirmed that the observed patterns persist under reasonable variations in pruning. In addition, we limit the number of maximum children for each feature to be 5. The following findings are subject to the limitations described in Section 7:

Main Takeaways

- **Hierarchical depth is layer-specific:** Concept trees vary in depth across layers, showing that only some layers exhibit strong hierarchical structure while others remain relatively flat. Thus, these findings suggest that the presence of hierarchical features is layer-dependent.
- **Compositionality is layer-specific:** The branching of concepts at the same hierarchy level varies across layers, with some layers exhibiting many children per concept while others have far fewer, indicating significant layer-specific differences in compositionality.
- **Consistency across models:** Both GPT-2-small and Gemma-2-2B (see Appendix A for width analysis of trees in Gemma) display the similar qualitative patterns (layer-selective structural properties), suggesting that layer-specific hierarchy and compositionality are general properties of transformer MLPs and may share similar trends across models.

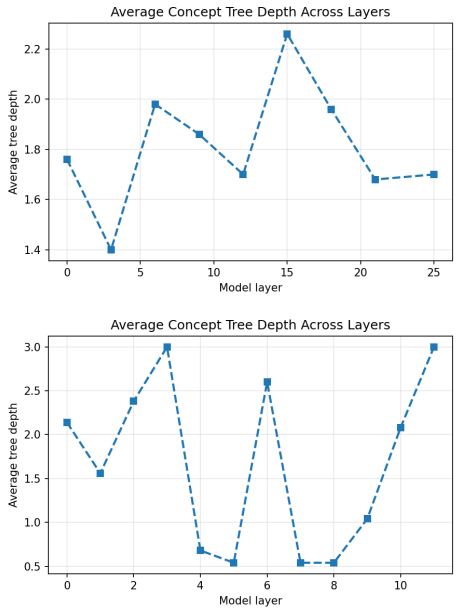

Figure 5: Average tree depth across layers. **Top:** Gemma-2-2B, **Bottom:** GPT-2-small. Some layers produce significantly deeper hierarchies, while others remain shallow, showing that hierarchical organization is layer-specific and consistent across models.

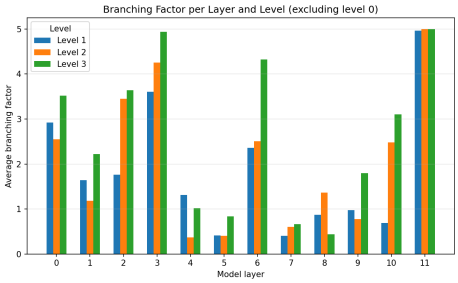

Figure 6: Average branching of concepts across levels and layers in GPT-2-small. Some layers exhibit wide, highly compositional structures, while others remain narrow, indicating that compositionality varies across layers.

6 Conclusions

Our investigation reveals distinct variations in steering efficacy across different levels of the SNMF hierarchy. Specifically, we observe a trend where lower hierarchy levels (representing more granular features) yield higher concept scores compared to deeper, more abstract levels. This deterioration in performance was persistent across both the 4-level decomposition baseline established by (Shafran et al., 2025) and our extended 5-level configuration with higher initial ranks. These results suggest that while hierarchical features at deeper levels are effective for identifying shared neural structures for complex concepts, as demonstrated by (Shafran et al., 2025), they may be inherently less "steerable" as isolated interpretable vectors.

Furthermore, we identified a phenomenon we term "hierarchical concept convergence," wherein higher hierarchy levels frequently produce nearly identical conceptual descriptions. While this convergence may partially stem from the LLM-based methodology used for feature identification, it also points toward potential structural limitations of SNMF decomposition when applied to specific rank-level configurations (see 7).

Finally, our structural analysis reveals substantial cross-layer variation in concept tree depth and width, shedding new light the structure of hierarchical concept representations obtained through SNMF decomposition.

7 Limitations and Future Work

Although our results highlight a decline in steering performance across the SNMF hierarchy, we cannot conclusively determine if this degradation is an intrinsic property of the hierarchical SNMF algorithm. The observed scores may be influenced

by confounding factors, including the composition of the concept dataset or the specific rank-to-level ratios selected for the decomposition.

Additionally, our evaluation framework, consistent with (Shafran et al., 2025), relies heavily on LLMs for both feature description and scoring. While this approach enables large-scale analysis, we found the LLM-based descriptions to be highly sensitive to prompt engineering. We have hypothesized potential biases of using LLMs as judges, particularly regarding functional "noise" concepts (e.g., punctuation and grammar), in section 4.1.

We suspect that the dataset's inherent hierarchical depth, the diversity of its descriptive sentences, and the chosen decomposition hyperparameters (number of levels and rank per level) could significantly impact the resulting feature space. Due to time and computational resources constrained, we were unable to further explore this parameter space. Future research could investigate how varying these hyperparameters affects steering patterns and whether more robust evaluation metrics can mitigate the "noise" observed in LLM-based scoring.

AI Disclosure

We used LLMs during our work, primarily for coding (mainly creating skeletons for code), for rephrasing our words in this work, as a clever search-engine for finding relevant tools from other papers that might help us, and sometimes just to bounce ideas off. Regardless of the usage, we always verified the LLM, and always made sure that every idea that we conveyed is our own.

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A Width Hierarchical Analysis for Gemma-2-2B

Figure 7 shows the average branching of concepts across all hierarchy levels and layers in Gemma-2-2B.

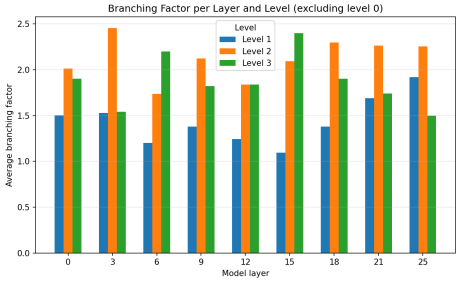


Figure 7: Average branching of concepts across hierarchy levels and layers in Gemma-2-2B. Some layers produce highly compositional concepts with many children, while others are narrower, showing strong layer-specific variation in compositionality.

Figure 8 focuses on level-1 concepts and shows that the same trend observed in GPT-2 also occurs in Gemma-2-2B. As in GPT-2, level-1 concepts become progressively more compositional from middle to later layers. This indicates that certain layers in the hierarchy produce wider, more compositional concepts, as representations move toward the deeper layers.

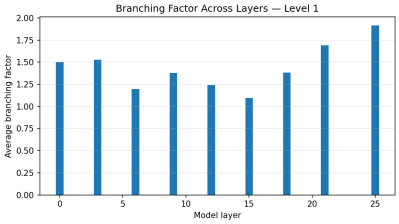


Figure 8: Average branching of level-1 concepts across layers in Gemma-2-2B. Middle to late layers produce more compositional level-1 concepts, consistent with the trends observed in GPT-2-small.

B Prompts

We used the following prompt for generating input descriptions based on top activating tokens. This is a modified version of the prompt used in (Shafran et al., 2025) as we used a different model (Gemini 2.5 Flash instead of GPT-4o) for generating the descriptions, and had to adjust the prompt.

Input Description Prompt

You are given a set of tokens, their surrounding context (words before and after the token), and an importance score. Your task is to identify a brief, direct concept label that describes the connection between these tokens.

Instructions:

1. **Focus on High-Importance Samples:**
   Examine only the token-context pairs with the highest importance scores. If a significant drop is observed beyond a threshold, ignore the lower-scoring pairs.

2. **Prioritize Tokens First:**
   - First look for patterns in the tokens themselves (identical tokens, related words, grammatical features, suffixes/prefixes, semantic patterns, etc.)
   - Only examine contexts if the tokens themselves show no clear relationship

3. **Output a Concise Label:**
   - Provide ONLY a brief concept label or short phrase
   - Do NOT start with phrases like "The tokens represent..." or "The concept is..."
   - If one specific token dominates, name it directly in quotes
   - Keep it as short as possible while being meaningful

Input:
Token-Context Pairs: {token_context_str}

Output only the concept label:

C Trends in hierarchical SNMF decomposition

As we inspected the results, we noticed some trends that immerge with the dataset and ranks that we chose for the hierarchical decomposition that may effect the steering concept scores.

C.1 Convergence of feature descriptions

We noticed that descriptions for concepts tends to be almost identical starting from hierarchy level 1. To investigate this, we measured, for each hierarchy level (except for the last one) the fraction of shared "concepts" (meaning, features with shared descriptions) with all the next levels. We did this for each model and each layer we analyzed the steering results for. The results for the 4-level decomposition are shown in Figure 9, Figure 10 and Figure 11. The results for the 5-level decomposition are shown in Figure 12 and Figure 13.

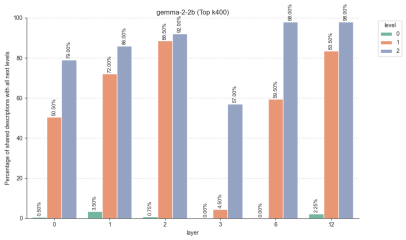


Figure 9: 4-level decomposition of Gemma-2-2B: Percentage of the shared concept descriptions each level has relative to all the next layers

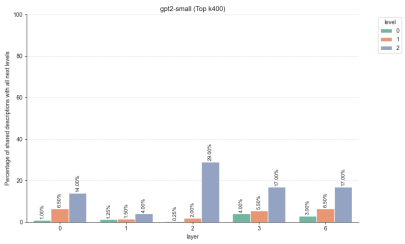


Figure 10: 4-level decomposition of GPT-2 Small: Percentage of the shared concept descriptions each level has relative to all the next layers

We see that the fraction of shared concepts increase with the level, and for most configurations, the fraction of shared concepts in levels 1 and above

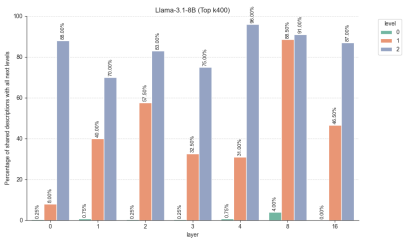


Figure 11: 4-level decomposition of Llama-3.1-8B: Percentage of the shared concept descriptions each level has relative to all the next layers

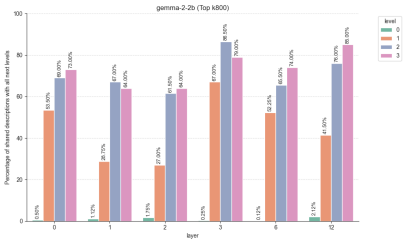


Figure 12: 5-level decomposition of Gemma-2-2B: Percentage of the shared concept descriptions each level has relative to all the next layers

is significant. We see that for the 4-level decomposition with GPT-2 small, the fraction of intersecting concept descriptions between levels 1,2,3 (shown in level 1) and between levels 2 and 3 is much lower compared to other models and also compared to GPT-2 Small with the 5-level decomposition. This also leads us to suspect that ideal ranks and datasets configurations are also dependent on the model.

C.2 Steering Score Trajectories in Concept Trees

Given the prevalence of noise identified in section 4.1 and the constraints discussed in section 7, we conducted a more granular analysis of steering concept score trajectories within the hierarchical trees. To facilitate this, we adapted the tree-construction methodology introduced by (Shafran et al., 2025) by employing a *top-p* selection strategy (Holtzman et al., 2020) to identify child nodes. Specifically, we set $p = 0.1$, effectively retaining the top 10% of the most highly activated child nodes for each parent node. We filtered out single-node trees as they are not relevant for this analysis. Each node in the resulting tree (representing a specific feature at a given hierarchy level) was then annotated with its corresponding LLM-generated input description and its empirical steering concept score. Our analysis focused exclusively on trees rooted at the highest hierarchical level (level

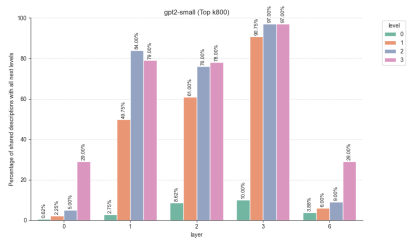
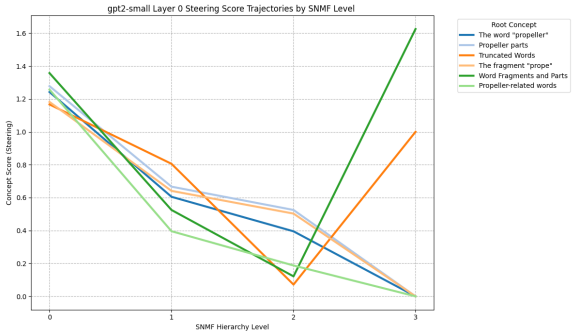


Figure 13: 5-level decomposition of GPT-2 Small: Percentage of the shared concept descriptions each level has relative to all the next layers

3 for the 4-level decomposition and level 4 for the 5-level decomposition).

We analyzed the steering score trajectories with respects to the concept that the root node represents. To account for the high degree of semantic overlap in the generated descriptions, we created a mapping to group related concepts. For instance, labels such as "Camera & Vehicle Parts," "Vehicle/Camera Parts & States," and "Vehicle/Camera Parts & Conditions" were unified under the aggregate category "Vehicle & Camera Combinations." The grouping was done by prompting Gemini-2.5 Flash Lite and Gemini 3 Pro, and then reorganized by code and verified manually.

This categorization rests on the assumption that roots with semantically similar concepts share overlapping nodes (features) and, consequently, should exhibit comparable steering score trajectories across hierarchy levels. For each aggregate category, we identified all relevant trees and calculated the mean concept score for each hierarchy level. The results are shown in Figure 14 - Figure 18. We can see that roots with domain-specific terms, such as "Optical instruments" (Figure 16) tend to have lower scores compared to starting scores. And higher scores in the last hierarchy leves are (mostly) attributed to root concepts that are morphosyntactic. There are some exceptions to this rule, and this is not a substantial quantitive analysis, but it offers points for considerations when investigating these steering score trajectories, such as finding ways to filter out this noise in order get a clearer conclusions in the steering analysis.



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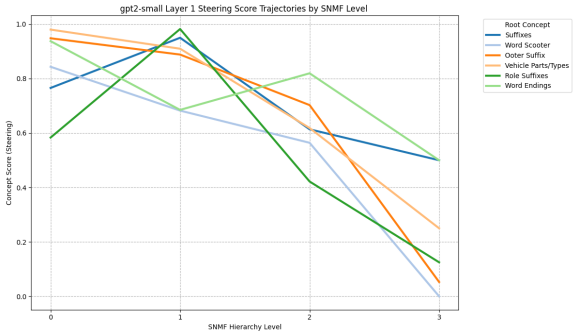

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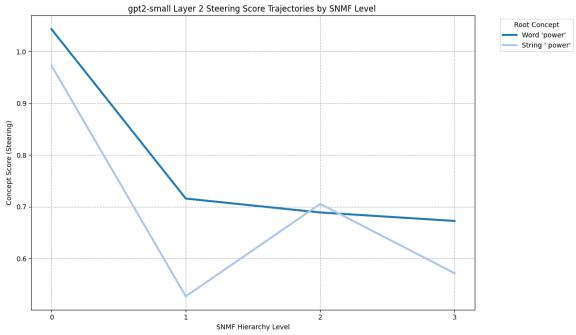
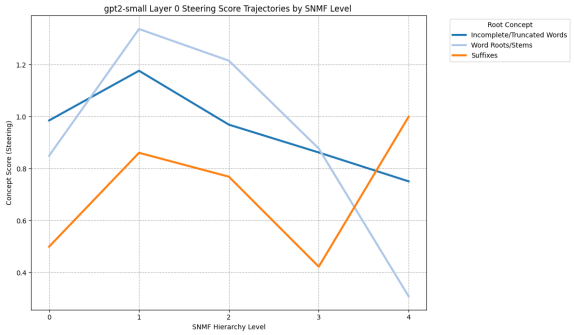


(c) Layer 2

Figure 14: Steering trajectories for the GPT-2 model in different layers with the 4-level hierarchical decomposition. These results show only 3 layers, because all other layers were filtered due to single-node trees.



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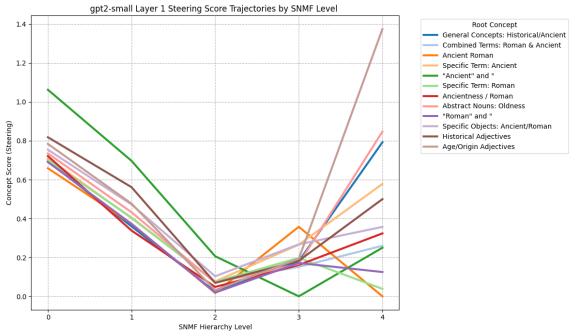

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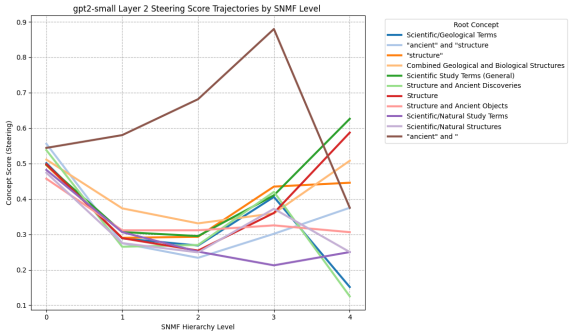

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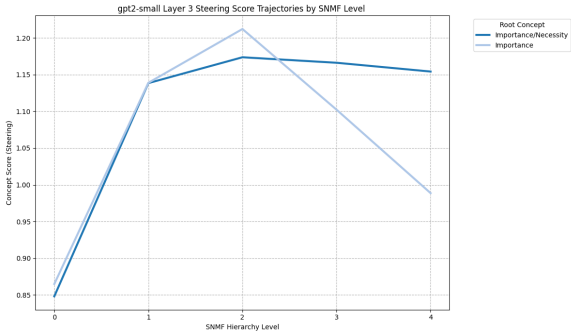

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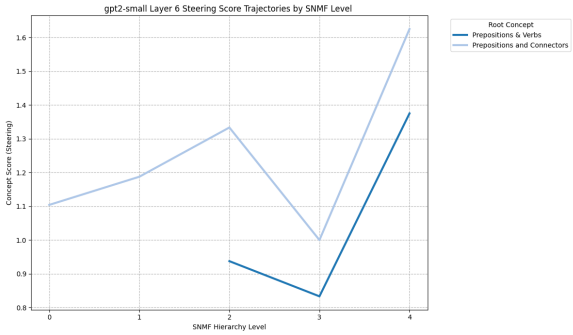
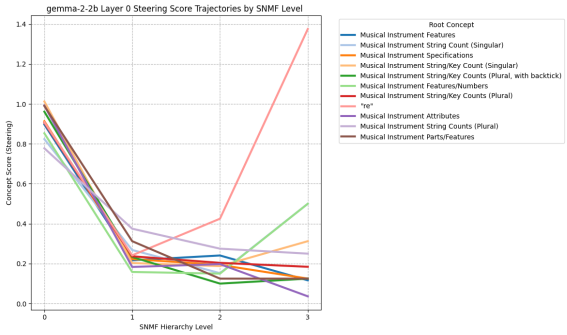


(e) Layer 6

Figure 15: Steering trajectories for the GPT-2 model in different layers with the 5-level hierarchical decomposition



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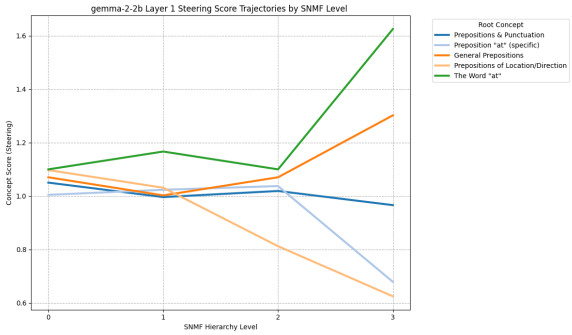

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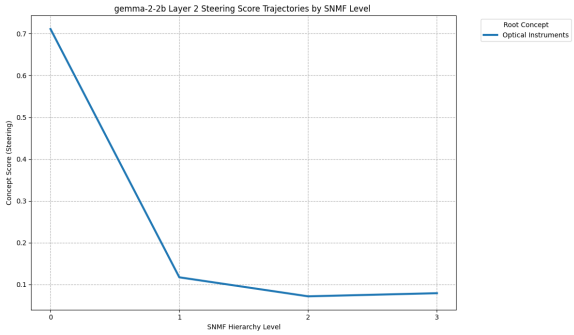

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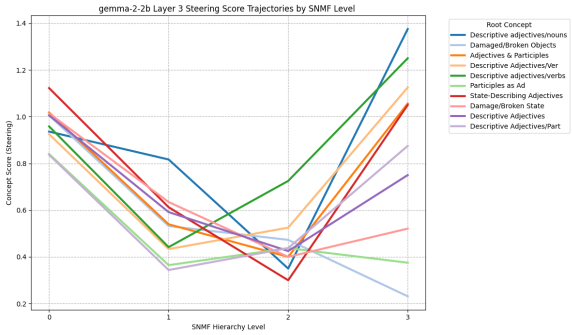

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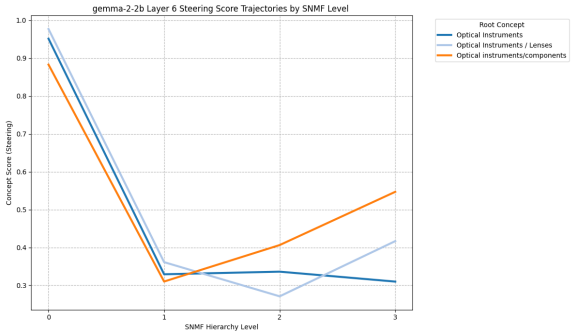

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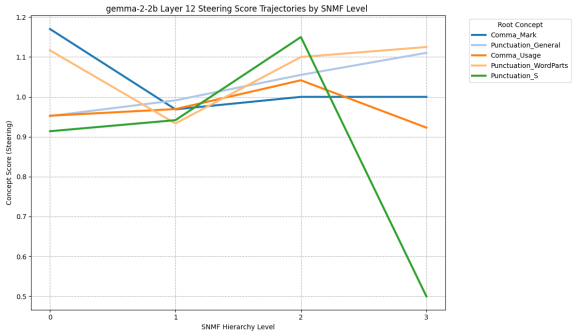
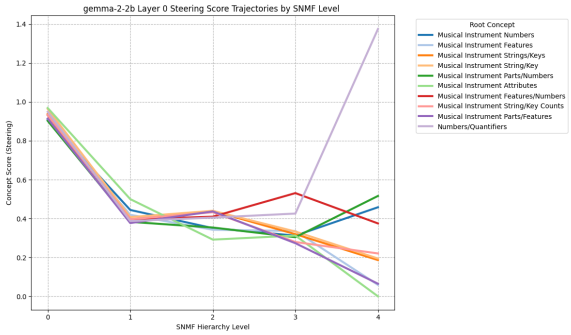


(f) Layer 12

Figure 16: Steering trajectories for Gemma-2-2b model in different layers with 4-level hierarchy decomposition.



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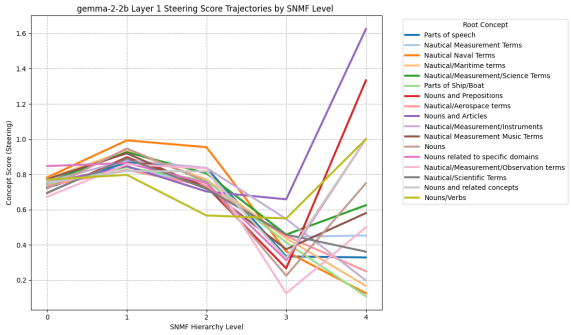

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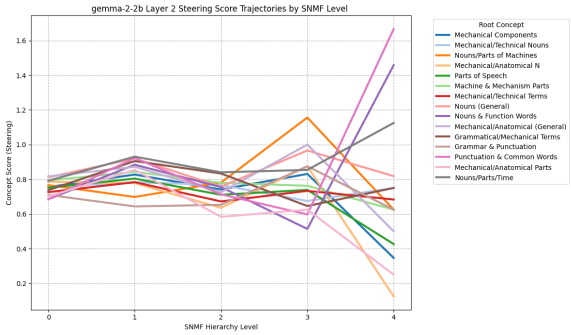

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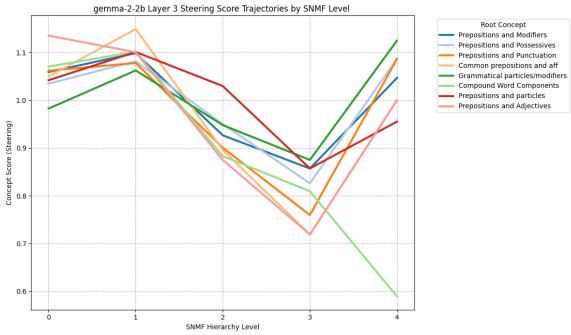

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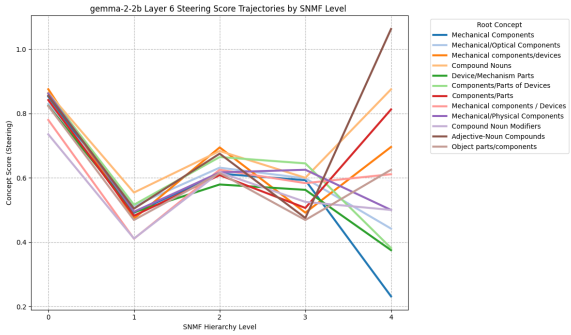

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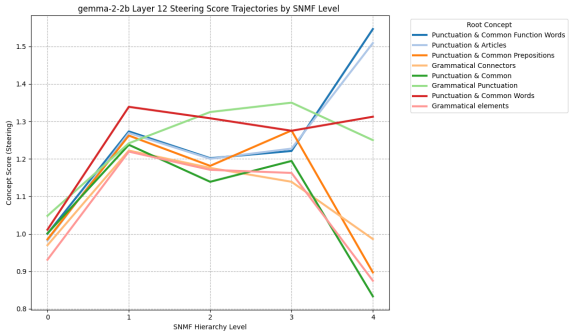
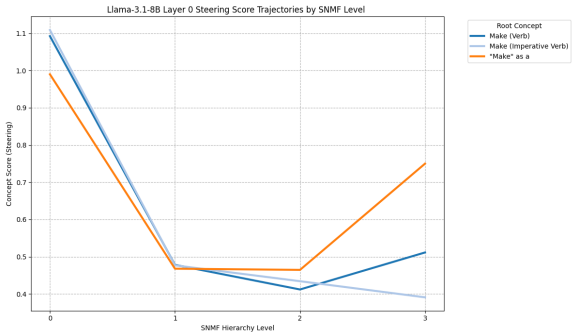


(f) Layer 12

Figure 17: Steering trajectories for Gemma-2-2b model in different layers with 5-level hierarchy decomposition.



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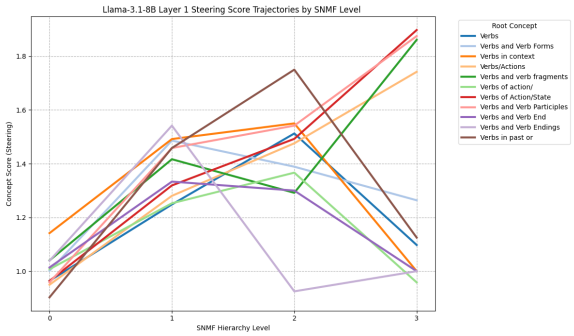

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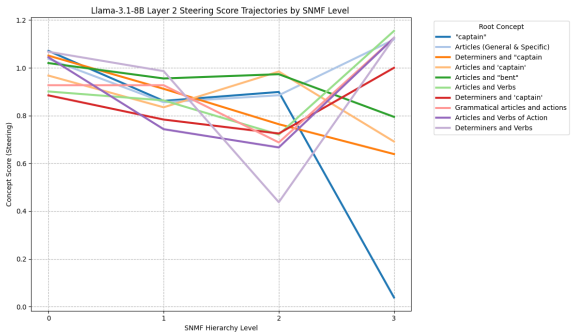

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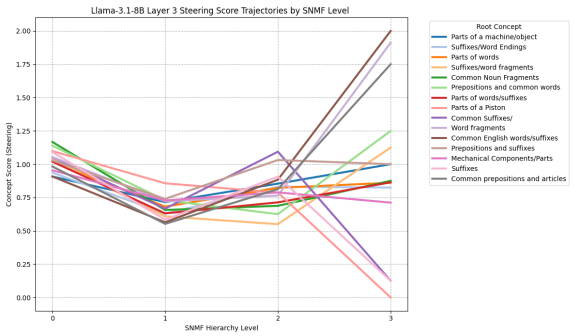

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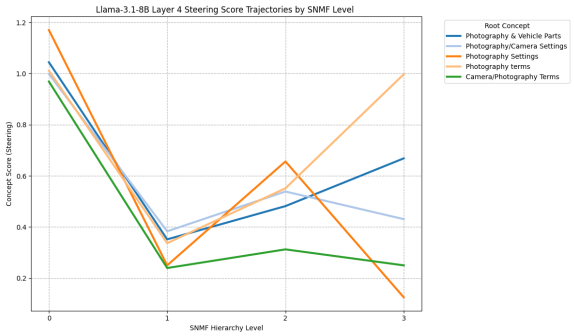

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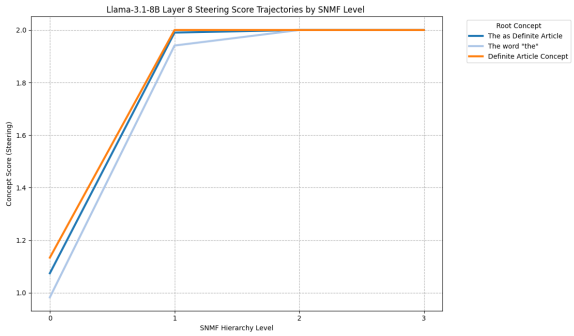

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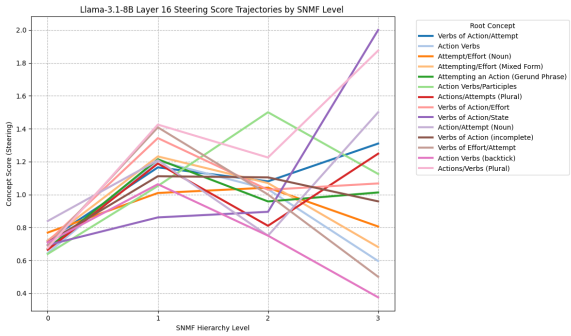

(g) Layer 16

Figure 18: Steering trajectories for Llama-3.1-8B model in different layers with 4-level hierarchy decomposition.